13 Parallel Clustering: Map-Reduce and Canopy Clustering

This chapter covers

- Understanding parallel and distributed computing
- Canopy clustering
- Parallelizing k-means by leveraging canopy clustering
- MapReduce computational model
- Using MapReduce to write a distributed version of k-means
- MapReduce canopy clustering
- MR-DBSCAN

In the previous chapter we have introduced clustering, and described three different approaches to data partitioning: k-means, DBSCAN, and OPTICS.

All these algorithms use a single-thread approach, where all the operations are executed sequentially in the same